

(A) AnchorScore Computation

Per-class images
+ domain prompts

CLIP ViT-L/14
zero-shot inference

AnchorScore
(per-class accuracy)

Apply τ to AnchorScore
(CLIP-predicted class)

High $\rightarrow$ CLIP
(cheap)

Low $\rightarrow$ MLLM
(expensive)

+1 to +23pp accuracy
44-77% cost saved

(i) Hybrid Annotation

Confusion
matrix

Enhanced prompts
(disambiguation)

MLLM accuracy
+25.0pp (direct)

(ii) Prompt Disambiguation

(B) Downstream Applications